

LOW STYRENE SBR PATENT LANDSCAPE: EMULSION, SOLUTION, AND FUNCTIONALIZED COPOLYMER ROUTES

1. ABSTRACT

This report synthesizes six primary patents relevant to low styrene styrene-butadiene rubber (SBR) and closely related styrene-butadiene copolymers. Across the manifest, the target is defined by reduced styrene fraction relative to conventional SBR, with disclosed embodiments spanning emulsion SBR, solution SBR, modified/functionalized SBR, and tapered or low-molecular-weight styrene-butadiene copolymers. The selected patents collectively emphasize control of styrene content, monomer feed ratio, sequence distribution, molecular weight, and post-polymerization functionalization as levers to tune tire-relevant properties such as wet grip, wear resistance, silica affinity, processability, and storage modulus.

Two major polymerization families dominate the landscape. Emulsion polymerization patents (US8530017B2, US9328176B2, EP2986647B1) use water as the continuous medium, with rosin/fatty-acid soaps or reactive emulsifiers, redox initiator systems, mercaptan chain-transfer agents, and low-temperature batch operation around 7–10 °C. These disclosures focus on introducing polar or reactive functionality during emulsion SBR synthesis, then using coagulation, washing, stripping, and drying to isolate the rubber. Solution polymerization patents (US11591428B2, US10174133B2, US11292862B2) use hydrocarbon solvents such as hexane, organolithium initiation, and controlled monomer feeds or terminal modification. These routes provide tighter control over styrene incorporation, microstructure, molecular weight, and functional end groups, while downstream water use is limited to coagulation or solvent-removal workup where disclosed.

A recurring formulation trend is the use of the resulting SBR in tire tread compounds with silica and/or carbon black, often alongside process oil, silane coupling agents, sulfur, and accelerators. Several patents explicitly target improved filler interaction or reduced oil demand, while maintaining or lowering styrene content into ranges consistent with low-styrene SBR embodiments. Overall, the patent set shows that low styrene SBR is not defined by a single synthesis route; rather, it is enabled by feed composition control and by functionalization strategies that preserve performance despite reduced styrene fraction.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US11591428B2
- Patent Title: Synthesis of low molecular weight tapered styrene-butadiene copolymer and its ...
- Assignee: Bridgestone Corporation
- Jurisdiction: US
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses solution polymerization of styrene and 1,3-butadiene in a hydrocarbon solvent using an organolithium initiator and sodium dodecylbenzenesulfonate, followed by coagulation in isopropanol and drum drying. The examples emphasize tapered styrene-butadiene copolymers with controlled sequence distribution and low molecular weight for use as oil replacements in rubber compounds. - Hexane: 0.18 kg — Example 1 - Styrene: 0.36 kg of 31.8 wt % in hexane — Example 1 - 1,3-Butadiene: 1.68 kg of 20.2 wt % in hexane — Example 1 - SDBS: 2.27 mL of 0.5 M in ethylbenzene — Example 1 - n-Butyl lithium: 6.05 mL of 1.60 M in hexane — Example 1 - Reaction temperature: 60° C. — Example 1 - Batch temperature peak: 69° C. — Example 1 - Polymerization time: 24 minutes + 2 hours — Example 1 - Coagulation medium: isopropanol containing butylated hydroxy toluene (BHT) — Example 1 - Mn: 52 kg/mol — Example 1 - Mw: 55 kg/mol — Example 1 - PDI: 1.05 — Example 1 - Tg: -76.07° C. — Example 1 - Percent styrene: 26.6% — Example 1 - Vinyl: 9.4% — Example 1 - 1,4-butadiene incorporation: 64% — Example 1 - Cis-microstructure: 39.4 weight percent of the butadiene units — Example 1 - Trans-microstructure: 49.6 weight percent of the butadiene units — Example 1 - Hexane: 0.30 kg — Example 2 - Styrene: 0.55 kg of 32.9 wt % in hexane — Example 2 - 1,3-Butadiene: 2.72 kg of 20 wt % in hexane — Example 2 - SDBS: 3.63 mL of 0.5 M in ethylbenzene — Example 2 - n-Butyl lithium: 9.07 mL of 1.60 M in hexane — Example 2 - Reaction temperature: 66° C. — Example 2 - Batch temperature peak: 107° C. — Example 2 - Polymerization time: 20 minutes + 40 minutes — Example 2 - Coagulation medium: isopropanol containing butylated hydroxy toluene (BHT) — Example 2 - Mn: 66 kg/mol — Example 2 - Mw: 74 kg/mol — Example 2 - PDI: 1.12 — Example 2 - Tg: -78.79° C. — Example 2 - Percent styrene: 25.7% — Example 2 - Vinyl: 9.7% — Example 2 - 1,4-butadiene incorporation: 64% — Example 2 - Cis-microstructure: 38.5 weight percent of the butadiene units — Example 2 - Trans-microstructure: 50.1 weight percent of the butadiene units — Example 2 - Medium & Water Role: Solution polymerization in hexane is the core reaction medium; the extracted evidence also discloses later coagulation in isopropanol, but no water-related process step is described. - low styrene fraction: 26.6% styrene (Example 1); 25.7% styrene (Example 2)

Medium & Water Role - Solution polymerization in hexane is the core reaction medium; the extracted evidence also discloses later coagulation in isopropanol, but no water-related process

step is described. - Core reaction medium: organic hydrocarbon solvent / solution polymerization in hexane

low styrene fraction - 26.6% styrene (Example 1); 25.7% styrene (Example 2)

Relevant Experimental Evidence - Example 1: Produced a tapered styrene-butadiene copolymer with 26.6% styrene and low PDI (1.05) in hexane solution polymerization. - Example 1: Demonstrated improved wear resistance when used as a partial or full oil replacement in carbon-black-filled rubber. - Example 2: Produced a similar tapered copolymer with 25.7% styrene and higher molecular weight than Example 1. - Example 2: Showed that the tapered copolymer improved wear index in rubber compositions relative to the oil-only control.

Technical Relevance This patent is directly relevant because it discloses a styrene-butadiene copolymer with styrene content in the low-to-moderate range and a solution process that enables control of sequence distribution and molecular weight. The disclosed copolymer is positioned for tire tread formulations, making it highly pertinent to low styrene SBR development.

Patent 2

- Patent Number: US8530017B2
- Patent Title: Method for preparing styrene-butadiene copolymer using reactive emulsifier and ...
- Assignee: Korea Kumho Petrochemical Co., Ltd.
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses radical emulsion polymerization of styrene and 1,3-butadiene in water using a reactive emulsifier that also participates in polymerization. The process uses a redox initiator system, mercaptan molecular-weight control, and low-temperature batch operation, followed by stripping, drying, and optional use in tire tread compounds. - Water: 1600 mL — Comparative Example 1 - Potassium rosinate: 22 g — Comparative Example 1 - Fatty acid potassium salt: 36 g — Comparative Example 1 - Potassium chloride: 1.5 g — Comparative Example 1 - Styrene: 290 g — Comparative Example 1 - 1,3-Butadiene: 710 g — Comparative Example 1 - EDTA: 0.5 g — Comparative Example 1 - Sodium hydrosulfite: 0.3 g — Comparative Example 1 - Ferrous sulfate: 0.1 g — Comparative Example 1 - Sodium formaldehyde sulfonate: 0.4 g — Comparative Example 1 - n-Dodecyl mercaptan: 1.0 g — Comparative Example 1 - Methane hydroperoxide: 0.6 g — Comparative Example 1 - Polymerization temperature: 10° C. — Comparative Example 1 - Polymerization time: 8 hours — Comparative Example 1 - Diethylhydroxyamine: 1.0 g — Comparative Example 1 - Weight average molecular weight: 920,000 g/mol — Comparative Example 1 - Water: 1600 mL — Example 1 - Potassium rosinate: 11 g — Example 1 - Fatty acid

potassium salt: 18 g — Example 1 - Potassium chloride: 1.5 g — Example 1 - Styrene: 260 g — Example 1 - Methylcarboxylate methacrylate: 30 g — Example 1 - 1,3-Butadiene: 710 g — Example 1 - EDTA: 0.5 g — Example 1 - Sodium hydrosulfite: 0.3 g — Example 1 - Ferrous sulfate: 0.1 g — Example 1 - Sodium formaldehyde sulfonate: 0.4 g — Example 1 - n-Dodecyl mercaptan: 1.0 g — Example 1 - Methane hydroperoxide: 0.6 g — Example 1 - Polymerization temperature: 10° C. — Example 1 - Polymerization time: 8 hours — Example 1 - Diethylhydroxyamine: 1.0 g — Example 1 - Weight average molecular weight: 860,000 g/mol — Example 1 - Water: 1600 mL — Example 2 - Potassium rosinate: 11 g — Example 2 - Fatty acid potassium salt: 18 g — Example 2 - Potassium chloride: 1.5 g — Example 2 - Styrene: 280 g — Example 2 - Polyoxyethylene phosphate methacrylate: 10 g — Example 2 - 1,3-Butadiene: 710 g — Example 2 - EDTA: 0.5 g — Example 2 - Sodium hydrosulfite: 0.3 g — Example 2 - Ferrous sulfate: 0.1 g — Example 2 - Sodium formaldehyde sulfonate: 0.4 g — Example 2 - n-Dodecyl mercaptan: 1.0 g — Example 2 - Methane hydroperoxide: 0.6 g — Example 2 - Polymerization temperature: 10° C. — Example 2 - Polymerization time: 8 hours — Example 2 - Diethylhydroxyamine: 1.0 g — Example 2 - Weight average molecular weight: 820,000 g/mol — Example 2 - Water: 1600 mL — Example 3 - Potassium rosinate: 11 g — Example 3 - Fatty acid potassium salt: 18 g — Example 3 - Potassium chloride: 1.5 g — Example 3 - Styrene: 260 g — Example 3 - Polyoxyethylene phosphate methacrylate: 30 g — Example 3 - 1,3-Butadiene: 710 g — Example 3 - EDTA: 0.5 g — Example 3 - Sodium hydrosulfite: 0.3 g — Example 3 - Ferrous sulfate: 0.1 g — Example 3 - Sodium formaldehyde sulfonate: 0.4 g — Example 3 - n-Dodecyl mercaptan: 1.0 g — Example 3 - Methane hydroperoxide: 0.6 g — Example 3 - Polymerization temperature: 10° C. — Example 3 - Polymerization time: 8 hours — Example 3 - Diethylhydroxyamine: 1.0 g — Example 3 - Weight average molecular weight: 824,000 g/mol — Example 3 - Water: 1600 mL — Example 4 - Potassium rosinate: 11 g — Example 4 - Fatty acid potassium salt: 18 g — Example 4 - Potassium chloride: 1.5 g — Example 4 - Styrene: 260 g — Example 4 - Allyl alcohol ethoxylate: 30 g — Example 4 - 1,3-Butadiene: 710 g — Example 4 - EDTA: 0.5 g — Example 4 - Sodium hydrosulfite: 0.3 g — Example 4 - Ferrous sulfate: 0.1 g — Example 4 - Sodium formaldehyde sulfonate: 0.4 g — Example 4 - n-Dodecyl mercaptan: 1.0 g — Example 4 - Methane hydroperoxide: 0.6 g — Example 4 - Polymerization temperature: 10° C. — Example 4 - Polymerization time: 8 hours — Example 4 - Diethylhydroxyamine: 1.0 g — Example 4 - Weight average molecular weight: 850,000 g/mol — Example 4 - Water: 1600 mL — Example 5 - Potassium rosinate: 11 g — Example 5 - Fatty acid potassium salt: 18 g — Example 5 - Potassium chloride: 1.5 g — Example 5 - Styrene: 260 g — Example 5 - Allyl alcohol polyethylene oxide phosphate: 30 g — Example 5 - 1,3-Butadiene: 710 g — Example 5 - EDTA: 0.5 g — Example 5 - Sodium hydrosulfite: 0.3 g — Example 5 - Ferrous sulfate: 0.1 g — Example 5 - Sodium formaldehyde sulfonate: 0.4 g — Example 5 - n-Dodecyl mercaptan: 1.0 g — Example 5 - Methane hydroperoxide: 0.6 g — Example 5 - Polymerization temperature: 10° C. — Example 5 - Polymerization time: 8 hours — Example 5 - Diethylhydroxyamine: 1.0 g — Example 5 - Weight average molecular weight: 900,000 g/mol — Example 5 - Water: 1600 mL — Example 6 - Potassium rosinate: 11 g — Example 6 - Fatty acid potassium salt: 18 g — Example 6 - Potassium chloride: 1.5 g — Example 6 - Styrene: 260 g — Example 6 - Allyl polyoxyethylene glycol methacrylate: 30 g — Example 6 - 1,3-Butadiene: 710 g — Example 6 - EDTA: 0.5 g — Example 6 - Sodium hydrosulfite: 0.3 g — Example 6 - Ferrous sulfate: 0.1 g — Example 6 - Sodium formaldehyde sulfonate: 0.4 g — Example 6 - n-Dodecyl mercaptan: 1.0 g — Example 6 -

Methane hydroperoxide: 0.6 g — Example 6 - Polymerization temperature: 10° C. — Example 6 - Polymerization time: 8 hours — Example 6 - Diethylhydroxyamine: 1.0 g — Example 6 - Weight average molecular weight: 950,000 g/mol — Example 6 - Water: 1600 mL — Example 7 - Potassium rosinate: 11 g — Example 7 - Fatty acid potassium salt: 18 g — Example 7 - Potassium chloride: 1.5 g — Example 7 - Styrene: 260 g — Example 7 - Polyoxypropylene glycol phosphate acrylate: 30 g — Example 7 - 1,3-Butadiene: 710 g — Example 7 - EDTA: 0.5 g — Example 7 - Sodium hydrosulfite: 0.3 g — Example 7 - Ferrous sulfate: 0.1 g — Example 7 - Sodium formaldehyde sulfonate: 0.4 g — Example 7 - n-Dodecyl mercaptan: 1.0 g — Example 7 - Methane hydroperoxide: 0.6 g — Example 7 - Polymerization temperature: 10° C. — Example 7 - Polymerization time: 8 hours — Example 7 - Diethylhydroxyamine: 1.0 g — Example 7 - Weight average molecular weight: 845,000 g/mol — Example 7 - Medium & Water Role: Water is the core polymerization medium in the emulsion/latex process, and it is also used later for coagulation and washing of the crumb rubber. - low styrene fraction: Not disclosed in extracted evidence

Medium & Water Role - Water is the core polymerization medium in the emulsion/latex process, and it is also used later for coagulation and washing of the crumb rubber. - Core reaction medium: aqueous emulsion / latex polymerization

low styrene fraction - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1: Prepared a standard SBR latex in water and converted it to dry rubber after coagulation, washing, and drying. - Example 1: Demonstrated a styrene-butadiene copolymer with 920,000 g/mol Mw under emulsion conditions. - Example 4: Introduced a reactive emulsifier monomer and obtained a functionalized SBR latex with 860,000 g/mol Mw. - Example 6: Showed that the functionalized copolymer could be extended with epoxidized soybean oil and still be processed into rubber compounds.

Technical Relevance This patent is highly relevant because it is a direct SBR emulsion-polymerization disclosure with styrene and butadiene feed ratios in the low-to-moderate styrene range and explicit control of emulsifier, initiator, and chain-transfer chemistry. It also demonstrates functionalized SBR variants that are used in tire tread formulations, which is important for low styrene SBR performance tuning.

Patent 3

- Patent Number: US10174133B2
- Patent Title: Modified styrene-butadiene copolymer, preparation method thereof, and rubber ...
- Assignee: Lg Chem, Ltd.
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses solution polymerization of styrene and 1,3-butadiene in normal hexane using n-butyllithium and a polar additive, followed by terminal capping with butadiene and post-polymerization modification with silane-containing modifiers. The polymer solution is then treated with ethanol, antioxidant, warm water, and roll drying to isolate the modified conjugated diene copolymer. - Styrene: 29.7 g — Example 1 - 1,3-Butadiene: 78.1 g — Example 1 - Normal hexane: 550 g — Example 1 - 2,2-di(2-tetrahydrofuryl)propane (DTP): 0.124 g — Example 1 - Reaction temperature: 50° C. — Example 1 - n-Butyllithium: 3 mmol — Example 1 - Additional 1,3-butadiene for capping: 2.2 g — Example 1 - Modifier: 0.181 g of 2-methoxyethyl 3-(bis(3-triethoxysilyl)propyl)amino)propanoate — Example 1 - Modifier/[act. Li]: 1.0 — Example 1 - DTP/[act. Li]: 2.3 — Example 1 - Ethanol: used to stop polymerization — Example 1 - BHT solution: 4.5 ml of 0.3 wt % in hexane — Example 1 - Styrene: 270 g — Example 2 - 1,3-Butadiene: 710 g — Example 2 - Normal hexane: 5 kg — Example 2 - 2,2-di(2-tetrahydrofuryl)propane (DTP): 0.94 g — Example 2 - Reaction temperature: 40° C. — Example 2 - n-Butyllithium: 25.40 g (2.62 wt % in hexane, 33% activation) — Example 2 - Additional 1,3-butadiene for capping: 20.0 g — Example 2 - Modifier: 1.98 g of an ethyl 3-(bis(3-triethoxy(methyl)silyl)propyl)amino)propanoate — Example 2 - Modifier/[act. Li]: 0.85 — Example 2 - DTP/[act. Li]: 1.46 — Example 2 - Ethanol: used to stop polymerization — Example 2 - Wingstay K solution: 33 g of 30 wt % in hexane — Example 2 - Weight average molecular weight: 120×10^4 g/mol — Example 1 (Table 1) - Number average molecular weight: 49×10^4 g/mol — Example 1 (Table 1) - Maximum peak molecular weight: 35×10^4 g/mol — Example 1 (Table 1) - Combination efficiency: 77% — Example 1 (Table 1) - Molecular weight distribution: 2.4 — Example 1 (Table 1) - Weight average molecular weight: 42×10^4 g/mol — Example 2 (Table 1) - Number average molecular weight: 27×10^4 g/mol — Example 2 (Table 1) - Maximum peak molecular weight: 30×10^4 g/mol — Example 2 (Table 1) - Combination efficiency: 64% — Example 2 (Table 1) - Molecular weight distribution: 1.4 — Example 2 (Table 1) - Medium & Water Role: The core polymerization medium is normal hexane solution; water appears later only in the warm-water treatment used to remove solvent before roll drying. - low styrene fraction: Not disclosed in extracted evidence

Medium & Water Role - The core polymerization medium is normal hexane solution; water appears later only in the warm-water treatment used to remove solvent before roll drying. - Core reaction medium: organic hydrocarbon solvent / solution polymerization in normal hexane

low styrene fraction - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1: Prepared a modified SSBR in hexane with DTP and a silane-containing modifier, then removed solvent with warm water and roll drying. - Example 2: Demonstrated a larger-scale solution polymerization and modification sequence with a different silane-containing modifier. - Example 1: Showed high combination efficiency and molecular weight for the modified polymer. - Example 2: Showed that the modified polymer could be produced with lower molecular weight distribution than Example 1.

Technical Relevance This patent is relevant because it is a direct solution SBR disclosure with explicit styrene and butadiene feed amounts and a controlled modification step. Although the

styrene fraction is not explicitly stated as a final composition in the extracted evidence, the patent clearly addresses styrene-butadiene copolymer synthesis and downstream functionalization relevant to low styrene SBR design.

Patent 4

- Patent Number: US9328176B2
- Patent Title: Functional styrene-butadiene copolymer
- Assignee: Korea Kumho Petrochemical Co., Ltd.
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses emulsion polymerization of styrene, butadiene, and an epoxy acrylate monomer in water, followed by acid- or base-catalyzed ring opening of the epoxy group to generate a functional styrene-butadiene copolymer. The resulting latex is compounded with silica and processed into tire tread sheets. - Water: 1,500 mL — Comparative Example 1 - Sodium rosinat: 25 g — Comparative Example 1 - Sodium salt of fatty acid: 35 g — Comparative Example 1 - Styrene: 400 g — Comparative Example 1 - 1,3-Butadiene: 600 g — Comparative Example 1 - EDTA: 0.6 g — Comparative Example 1 - Ferrous sulfate: 0.1 g — Comparative Example 1 - n-Dodecylmercaptan: 1.5 g — Comparative Example 1 - p-Menthane hydroperoxide: 0.6 g — Comparative Example 1 - Polymerization temperature: 10° C. — Comparative Example 1 - Diethylhydroxylamine: 1.0 g — Comparative Example 1 - Average molecular weight: 843,000 g/mol — Comparative Example 1 - Water: 1,500 mL — Comparative Example 2 - Sodium rosinat: 25 g — Comparative Example 2 - Sodium salt of fatty acid: 35 g — Comparative Example 2 - Styrene: 380 g — Comparative Example 2 - 1,3-Butadiene: 570 g — Comparative Example 2 - Methyl methacrylic acid: 50 g — Comparative Example 2 - EDTA: 0.6 g — Comparative Example 2 - Ferrous sulfate: 0.1 g — Comparative Example 2 - n-Dodecylmercaptan: 1.5 g — Comparative Example 2 - p-Menthane hydroperoxide: 0.6 g — Comparative Example 2 - Polymerization temperature: 10° C. — Comparative Example 2 - Diethylhydroxylamine: 1.0 g — Comparative Example 2 - Average molecular weight: 795,000 g/mol — Comparative Example 2 - Water: 1,500 mL — Example 1 — Example 1 - Sodium rosinat: 25 g — Example 1 - Sodium salt of fatty acid: 35 g — Example 1 - Styrene: 380 g — Example 1 - Glycidyl methyl methacrylate: 50 g — Example 1 - 1,3-Butadiene: 570 g — Example 1 - EDTA: 0.6 g — Example 1 - Ferrous sulfate: 0.1 g — Example 1 - n-Dodecylmercaptan: 1.5 g — Example 1 - p-Menthane hydroperoxide: 0.6 g — Example 1 - Polymerization temperature: 10° C. — Example 1 - Polymerization time: 24 hours — Example 1 - Diethylhydroxylamine: 1.0 g — Example 1 - Sulfuric acid: 20% aqueous solution, 100 g — Example 1 - Average molecular weight: 908,000 g/mol — Example 1 - Water: 1,500 mL — Example 4 - Sodium rosinat: 25 g — Example 4 - Sodium salt of fatty acid: 35 g — Example 4 - Styrene: 380 g — Example 4 - 1,3-Butadiene: 570 g — Example 4 - EDTA: 0.6 g — Example 4 -

Ferrous sulfate: 0.1 g — Example 4 - n-Dodecylmercaptan: 1.5 g — Example 4 - p-Menthane hydroperoxide: 0.6 g — Example 4 - Polymerization temperature: 10° C. — Example 4 - Diethylhydroxylamine: 1.0 g — Example 4 - NaOH: 20% aqueous solution, 100 g — Example 4 - Average molecular weight: 740,000 g/mol — Example 4 - Water: 1,500 mL — Example 5 - Sodium rosinate: 25 g — Example 5 - Sodium salt of fatty acid: 35 g — Example 5 - Styrene: 380 g — Example 5 - 1,3-Butadiene: 570 g — Example 5 - EDTA: 0.6 g — Example 5 - Ferrous sulfate: 0.1 g — Example 5 - n-Dodecylmercaptan: 1.5 g — Example 5 - p-Menthane hydroperoxide: 0.6 g — Example 5 - Polymerization temperature: 10° C. — Example 5 - Diethylhydroxylamine: 1.0 g — Example 5 - KOH: 20% aqueous solution, 100 g — Example 5 - Average molecular weight: 791,000 g/mol — Example 5 - Medium & Water Role: Water is the polymerization medium in the emulsion process, and aqueous acid/base solutions are later used for epoxy ring opening and latex workup. - low styrene fraction: Not disclosed in extracted evidence

Medium & Water Role - Water is the polymerization medium in the emulsion process, and aqueous acid/base solutions are later used for epoxy ring opening and latex workup. - Core reaction medium: aqueous emulsion / latex polymerization

low styrene fraction - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1: Prepared a styrene-butadiene-dihydroxypropyl methacrylate copolymer by emulsion polymerization and sulfuric-acid ring opening. - Example 4: Showed that NaOH can be used as the ring-opening agent to obtain the hydroxyl-functional copolymer. - Example 5: Showed that KOH can be used as the ring-opening agent to obtain the hydroxyl-functional copolymer. - Comparative Example 1: Provided the unfunctionalized SBR baseline for comparison.

Technical Relevance This patent is relevant because it is a direct SBR emulsion-polymerization disclosure with styrene and butadiene feed ratios that can support low-styrene embodiments, while also introducing polar functionality for silica compatibility. The work is important for low styrene SBR because it addresses how to preserve tire-relevant performance while modifying the copolymer architecture.

Patent 5

- Patent Number: US11292862B2
- Patent Title: Modified conjugated diene-based polymer, polymer composition, and rubber ...
- Assignee: Asahi Kasei Kabushiki Kaisha
- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses continuous solution polymerization of purified 1,3-butadiene and styrene in n-hexane using n-butyllithium and a polar substance, followed by in-line modification with silane-containing modifiers and steam stripping to remove solvent. The examples vary monomer feed rates, initiator rate, polar additive rate, and modifier rate to tune molecular weight, styrene incorporation, and modification ratio. - Water washing step: circulating water 1 m³/hr — Purification of 1,3-butadiene - Makeup water: 0.1 m³/hr — Purification of 1,3-butadiene - Liquid temperature: 30° C. — Water washing step - Decanter pressure: 1.0 MPaG — Water washing step - Residence time of 1,3-butadiene phase: 30 minutes — Water washing step - Steam heating: 89° C. — Water washing step - Total pressure: 0.01 MPaG — Water washing step - 10% aqueous solution of Diclean F-504 — Oxygen removal step - Residence time of 1,3-butadiene phase: 30 minutes — Oxygen removal step - 10% aqueous solution of caustic soda — Polymerization inhibitor removal step - Residence time of 1,3-butadiene phase: 60 minutes — Polymerization inhibitor removal step - Mixed hexane concentration of 1,3-butadiene: 50% by mass — Dehydrating column step - Styrene feed rate: 9.9 g/min — Table 1 - 1,3-Butadiene feed rate: 18.1 g/min — Table 1 - Normal hexane feed rate: 150.1 g/min — Table 1 - Total impurities: 23 ppm — Table 1 - Polymerization temperature: 70° C. — Table 1 - n-Butyllithium for treatment: 0.104 mmol/min — Table 1 - n-Butyllithium as polymerization initiator: 0.252 mmol/min — Table 1 - Polar substance: 0.0216 g/min — Table 1 - Modifier amount: 0.043 mmol/min — Table 1 - Bound styrene: 35% by mass — Table 1 - 1,2-Vinyl bond: 41 mol % — Table 1 - Weight average molecular weight: 65.1 ×10⁴ g/mol — Table 1 - Number average molecular weight: 31.8 ×10⁴ g/mol — Table 1 - Peak top molecular weight: 63.6 ×10⁴ g/mol — Table 1 - Molecular weight distribution: 2.05 — Table 1 - Ratio of polymer having molecular weight of 2 to 5 million: 1.6% by mass — Table 1 - Mooney viscosity of polymer (100° C.): 56 — Table 1 - Glass transition temperature: -24° C. — Table 1 - Modification ratio: 81% by mass — Table 1 - Modification ratio of low molecular weight component: 47% by mass — Table 1 - Modification degree of low molecular weight component: 58% — Table 1 - Shrinkage factor (g'): 0.68 — Table 1 - Silicon content: 57 mass ppm — Table 1 - Nitrogen content: 19 mass ppm — Table 1 - Nitrogen/Silicon molar ratio: 0.67 — Table 1 - Styrene feed rate: 5 g/min — Table 1 (Sample 7) - 1,3-Butadiene feed rate: 23 g/min — Table 1 (Sample 7) - Polar substance: 0.0155 g/min — Table 1 (Sample 7) - n-Butyllithium as polymerization initiator: 0.252 mmol/min — Table 1 (Sample 7) - Modifier amount: 0.043 mmol/min — Table 1 (Sample 7) - Bound styrene: 17% by mass — Table 1 (Sample 7) - 1,2-Vinyl bond: 23 mol % — Table 1 (Sample 7) - Weight average molecular weight: 64.9 ×10⁴ g/mol — Table 1 (Sample 7) - Number average molecular weight: 30.1 ×10⁴ g/mol — Table 1 (Sample 7) - Peak top molecular weight: 70.1 ×10⁴ g/mol — Table 1 (Sample 7) - Molecular weight distribution: 2.16 — Table 1 (Sample 7) - Ratio of polymer having molecular weight of 2 to 5 million: 1.5% by mass — Table 1 (Sample 7) - Mooney viscosity of polymer (100° C.): 54 — Table 1 (Sample 7) - Glass transition temperature: -63° C. — Table 1 (Sample 7) - Styrene feed rate: 12 g/min — Table 1 (Sample 8) - 1,3-Butadiene feed rate: 16 g/min — Table 1 (Sample 8) - Polar substance: 0.024 g/min — Table 1 (Sample 8) - n-Butyllithium as polymerization initiator: 0.252 mmol/min — Table 1 (Sample 8) - Modifier amount: 0.033 mmol/min — Table 1 (Sample 8) - Bound styrene: 42% by mass — Table 1 (Sample 8) - 1,2-Vinyl bond: 46 mol % — Table 1 (Sample 8) - Weight average molecular weight: 81.3 ×10⁴ g/mol — Table 1 (Sample 8) - Number average molecular weight: 37.8 ×10⁴ g/mol — Table 1 (Sample 8) - Peak top molecular weight: 88.3 ×10⁴ g/mol — Table 1 (Sample 8) - Molecular weight

distribution: 2.15 — Table 1 (Sample 8) - Ratio of polymer having molecular weight of 2 to 5 million: 4% by mass — Table 1 (Sample 8) - Mooney viscosity of polymer (100° C.): 64 — Table 1 (Sample 8) - Glass transition temperature: -10° C. — Table 1 (Sample 8) - Styrene feed rate: 9.9 g/min — Table 1 (Sample 9) - 1,3-Butadiene feed rate: 18.1 g/min — Table 1 (Sample 9) - Polar substance: 0.0076 g/min — Table 1 (Sample 9) - n-Butyllithium as polymerization initiator: 0.08 mmol/min — Table 1 (Sample 9) - Modifier amount: 0.03 mmol/min — Table 1 (Sample 9) - Bound styrene: 35% by mass — Table 1 (Sample 9) - 1,2-Vinyl bond: 41 mol % — Table 1 (Sample 9) - Weight average molecular weight: 31.2×10^4 g/mol — Table 1 (Sample 9) - Number average molecular weight: 15.0×10^4 g/mol — Table 1 (Sample 9) - Peak top molecular weight: 29.1×10^4 g/mol — Table 1 (Sample 9) - Molecular weight distribution: 2.08 — Table 1 (Sample 9) - Ratio of polymer having molecular weight of 2 to 5 million: 0.6% by mass — Table 1 (Sample 9) - Mooney viscosity of polymer (100° C.): 31 — Table 1 (Sample 9) - Glass transition temperature: -24° C. — Table 1 (Sample 9) - Medium & Water Role: The core polymerization medium is n-hexane solution; water is used upstream in monomer purification and downstream in washing/dehydration and steam-stripping-related workup steps. - low styrene fraction: 17% by mass bound styrene — Table 1 (Sample 7)

Medium & Water Role - The core polymerization medium is n-hexane solution; water is used upstream in monomer purification and downstream in washing/dehydration and steam-stripping-related workup steps. - Core reaction medium: organic hydrocarbon solvent / continuous solution polymerization in n-hexane

low styrene fraction - 17% by mass bound styrene — Table 1 (Sample 7)

Relevant Experimental Evidence - Example 1: Established continuous solution polymerization with purified feeds, in-line modification, and steam stripping to obtain a modified conjugated diene-based polymer. - Example 1: Achieved 35% bound styrene and 81% modification ratio in the main sample set. - Sample 7: Lower styrene feed rate gave 17% bound styrene and a much lower Tg of -63° C. - Sample 8: Higher styrene feed rate gave 42% bound styrene and a higher Tg of -10° C. - Sample 9: Lower initiator and modifier rates produced a lower-Mw polymer with 35% bound styrene.

Technical Relevance This patent is highly relevant because it directly discloses solution SBR synthesis with explicit control of bound styrene content, including a low-styrene example at 17% by mass. It also links monomer feed ratio and polymerization conditions to properties important for low styrene SBR, such as Tg, Mooney viscosity, and modification ratio.

Patent 6

- Patent Number: EP2986647B1
- Patent Title: Oil extended functionalized styrene-butadiene copolymer
- Assignee: ARLANXEO Deutschland GmbH
- Jurisdiction: EP
- Publication Year: 2017

- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses emulsion polymerization of styrene and butadiene in water to make standard and functionalized SBR latex, followed by blending with extender oil, coagulation with steam heating, washing with demineralized water, and drying. A functionalized variant is prepared by adding glycidyl methacrylate at the start of polymerization and then using the latex in oil-extended rubber formulations. - Water: 8,700 g — Example 1 - Emulsifier batch with potassium salt of fatty and rosin acids: 2,330 g — Example 1 - 1,3-Butadiene: 3,830 g — Example 1 - Styrene: 1,486 g — Example 1 - Pinane hydroperoxide: 2.5 g — Example 1 - Activator batch with water, ferrous sulfate, EDTA and sodium formaldehyde sulfoxylate: 390 g — Example 1 - t-Dodecyl mercaptan: 10 g — Example 1 - Polymerization temperature: 7°C — Example 1 - Conversion: 60 % — Example 1 - Polymerization duration: about 7 hours — Example 1 - Oil amount in dry rubber: 37.5 phr — Example 1 - Coagulation temperature: 65°C — Example 1 - Drying time: 18 hours — Example 1 - Water: 8,500 g — Example 4 - Emulsifier batch with potassium salt of fatty and rosin acids: 2,330 g — Example 4 - Phosphoric acid 85 %: 2.1 g — Example 4 - Acetic acid 99 %: 5.4 g — Example 4 - Potassium hydroxide 50 %: 15.3 g — Example 4 - Water in buffering system: 200.0 g — Example 4 - pH: 8.7 — Example 4 - 1,3-Butadiene: 3,830 g — Example 4 - Styrene: 1,242 g — Example 4 - Blend of GMA/styrene: 106/133 g — Example 4 - Pinane hydroperoxide: 2.5 g — Example 4 - Activator batch with water, ferrous sulfate, EDTA and sodium formaldehyde sulfoxylate: 390 g — Example 4 - t-Dodecyl mercaptan: 10 g — Example 4 - Polymerization temperature: 7°C — Example 4 - Conversion: 60 % — Example 4 - Polymerization duration: about 5 hours — Example 4 - pH of final latex: 9.2 — Example 4 - Oil amount in dry rubber: 37.5 phr — Example 4 - Coagulation temperature: 65°C — Example 4 - Drying time: 18 hours — Example 4 - Bound styrene content: 21.0 %wt — Example 6 - cis-1.4: 9.5 %wt — Example 6 - trans-1.4: 76.3 %wt — Example 6 - vinyl 1.2: 14.2 %wt — Example 6 - Mooney viscosity (ML 1+4, 100°C): 45.8 — Example 6 - Average molecular weight (Mw): 356,000 g/mol — Example 6 - Rubber in compound: 96.2 phr — Table 2 and Table 3 - Buna CB 24 (polybutadiene): 30.0 phr — Table 2 and Table 3 - Carbon black (N-339): 75.0 phr — Table 2 - Oil (naphthenic, common soybean or epoxidized soybean): 11.3 phr — Table 2 and Table 3 - Silica (Zeozil 1165 MP): 70.0 phr — Table 3 - Silane (Si 69): 5.6 phr — Table 3 - Medium & Water Role: Water is the polymerization medium in the emulsion latex process, and it is also used later for coagulation and washing of the crumb rubber. - low styrene fraction: 21.0 %wt bound styrene — Example 6

Medium & Water Role - Water is the polymerization medium in the emulsion latex process, and it is also used later for coagulation and washing of the crumb rubber. - Core reaction medium: aqueous emulsion / latex polymerization

low styrene fraction - 21.0 %wt bound styrene — Example 6

Relevant Experimental Evidence - Example 1: Prepared standard SBR latex by emulsion polymerization and converted it to dry rubber after oil blending, coagulation, washing, and drying. -

Example 4: Prepared functionalized SBR latex by adding GMA at the start of emulsion polymerization and achieved final latex pH 9.2. - Example 6: Reported a functionalized SBR with 21.0 wt% bound styrene and Mw of 356,000 g/mol. - Example 6: Demonstrated use of the functionalized SBR in carbon-black and silica compounds with vegetable-oil-based extender oils.

Technical Relevance This patent is relevant because it is a direct SBR emulsion-polymerization disclosure and explicitly reports a functionalized SBR with 21.0 wt% bound styrene, which falls within a low-styrene regime relative to conventional grades. It also emphasizes oil extension and silica compatibility, both important for low styrene SBR applications in tire tread compounds.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	low styrene fraction	Key Findings
US11591428B2	Bridgestone Corporation	2023	Solution polymerization in hexane is the core reaction medium; the extracted evidence also discloses later coagulation in isopropanol, but no water-related process step is described.	26.6% styrene (Example 1); 25.7% styrene (Example 2)	Example 1: Produced a tapered styrene-butadiene copolymer with 26.6% styrene and low PDI (1.05) in hexane solution polym

US8530017B2	Korea Kumho Petrochemical Co., Ltd.	2013	Water is the core polymerization medium in the emulsion/latex process, and it is also used later for coagulation and washing of the crumb rubber.	Not disclosed in extracted evidence	Example 1: Prepared a standard SBR latex in water and converted it to dry rubber after coagulation, washing, and drying.
US10174133B2	Lg Chem, Ltd.	2019	The core polymerization medium is normal hexane solution; water appears later only in the warm-water treatment used to remove solvent before roll drying.	Not disclosed in extracted evidence	Example 1: Prepared a modified SSBR in hexane with DTP and a silane-containing modifier, then removed solvent with warm

US9328176B2	Korea Kumho Petrochemical Co., Ltd.	2016	Water is the polymerization medium in the emulsion process, and aqueous acid/base solutions are later used for epoxy ring opening and latex workup.	Not disclosed in extracted evidence	Example 1: Prepared a styrene-butadiene-dihydro methacrylate copolymer by emulsion polymerization and sulfuric-a
US11292862B2	Asahi Kasei Kabushiki Kaisha	2022	The core polymerization medium is n-hexane solution; water is used upstream in monomer purification and downstream in washing/dehydration and steam-stripping-related workup steps.	17% by mass bound styrene — Table 1 (Sample 7)	Example 1: Established continuous solution polymerization with purified feeds, in-line modification, and steam stripping

EP2986647B1	ARLANXEO Deutschland GmbH	2017	Water is the polymerization medium in the emulsion latex process, and it is also used later for coagulation and washing of the crumb rubber.	21.0 %wt bound styrene — Example 6	Example 1: Prepared standard SBR latex by emulsion polymerization and converted it to dry rubber after oil blending, coa
-------------	---------------------------------	------	--	---------------------------------------	--

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US11591428B2 and US10174133B2 are solution-polymerized SBR-type materials in hydrocarbon solvent, whereas US8530017B2, US9328176B2, and EP2986647B1 are emulsion/latex processes in water.
- The lowest explicitly disclosed styrene fractions in the manifest are 17% bound styrene in US11292862B2 Sample 7 and 21.0 wt% bound styrene in EP2986647B1 Example 6; US11591428B2 reports 25.7-26.6% styrene in tapered copolymers.
- US11292862B2 uniquely combines continuous solution polymerization with in-line purification of monomers, modifier addition, and steam stripping, while US10174133B2 emphasizes terminal capping and silane-based post-modification.
- US8530017B2 and US9328176B2 both use water-based emulsion polymerization with redox initiators and mercaptan chain-transfer agents, but US8530017B2 focuses on reactive emulsifiers, whereas US9328176B2 focuses on epoxy acrylate functionalization and ring opening.
- EP2986647B1 differs from the other emulsion patents by explicitly incorporating vegetable-oil-based extender oils and reporting oil-extended functionalized SBR latex and dry rubber properties.
- Across the manifest, styrene content is controlled either by monomer feed ratio in solution polymerization or by emulsion recipe design; functionalization is used to recover silica affinity, processability, and tread performance at reduced styrene levels.

4. CONCLUSION

The selected primary patents show two complementary routes to low styrene SBR: water-based emulsion polymerization with functional or reactive emulsifiers, and hydrocarbon-solution polymerization with precise feed control and post-polymerization modification. The strongest direct low-styrene disclosures are US11292862B2 (17% bound styrene in a solution polymerized modified conjugated diene polymer) and EP2986647B1 (21.0 wt% bound styrene in a functionalized oil-extended SBR), while US11591428B2 provides tapered SBR at 25.7-26.6% styrene and the emulsion patents demonstrate how polar functionality and oil extension can preserve tire performance. Overall, the manifest indicates that low styrene fraction is achieved not by a single recipe but by coordinated control of monomer ratio, sequence distribution, molecular weight, and functionalization/workup strategy.

5. REFERENCES

- US11591428B2 | Synthesis of low molecular weight tapered styrene-butadiene copolymer and its ... | Bridgestone Corporation | US | 2023 | <https://patents.google.com/patent/US11591428B2/en>
- US8530017B2 | Method for preparing styrene-butadiene copolymer using reactive emulsifier and ... | Korea Kumho Petrochemical Co., Ltd. | US | 2013 | <https://patents.google.com/patent/US8530017B2/en>
- US10174133B2 | Modified styrene-butadiene copolymer, preparation method thereof, and rubber ... | Lg Chem, Ltd. | US | 2019 | <https://patents.google.com/patent/US10174133B2/en>
- US9328176B2 | Functional styrene-butadiene copolymer | Korea Kumho Petrochemical Co., Ltd. | US | 2016 | <https://patents.google.com/patent/US9328176B2/en>
- US11292862B2 | Modified conjugated diene-based polymer, polymer composition, and rubber ... | Asahi Kasei Kabushiki Kaisha | US | 2022 | <https://patents.google.com/patent/US11292862B2/en>
- EP2986647B1 | Oil extended functionalized styrene-butadiene copolymer | ARLANXEO Deutschland GmbH | EP | 2017 | <https://patents.google.com/patent/EP2986647B1/en>